

Pre-Algebra Practice Test I - Version One

Simplify the following expressions.

- ① a) $2 - 6 = \boxed{-4}$ b) $-5 + 10 = \boxed{5}$
 c) $(-7) + (-1) = \boxed{-8}$ d) $-8 + 7 = \boxed{-1}$
 ② a) $-3 + (-2) = \boxed{-5}$ b) $-4 - 7 = \boxed{-11}$
 c) $8 - (-5) = 8 + 5 = \boxed{13}$ d) $-6 - (-9) = -6 + 9 = \boxed{3}$
 ③ a) $3(-6) = \boxed{-18}$ b) $-2(-5) = \boxed{10}$
 b) $(-7)(-8) = \boxed{56}$ c) $(8)(-1) = \boxed{-8}$
 ④ a) $-5(2) = \boxed{-10}$ b) $-3(3) = \boxed{-9}$
 c) $(-10)7 = \boxed{-70}$ d) $(-11)(9) = \boxed{-99}$
 ⑤ a) $\frac{-49}{7} = \boxed{-7}$ b) $\frac{72}{-8} = \boxed{-9}$
 c) $\frac{-48}{-6} = \boxed{8}$ d) $-\frac{54}{9} = \boxed{-6}$
 ⑥ a) $\frac{-36}{60} = -\frac{6}{10} = \boxed{-\frac{3}{5}}$ b) $\frac{15}{-50} = \boxed{-\frac{3}{10}}$
 c) $\frac{-24}{-36} = \frac{4}{6} = \boxed{\frac{2}{3}}$ d) $-\frac{18}{22} = \boxed{-\frac{9}{11}}$

⑦ a) $\frac{1}{3} - \frac{5}{6} = \frac{2}{6} - \frac{5}{6} = -\frac{3}{6} = \boxed{-\frac{1}{2}}$ b) $-\frac{5}{6} + \frac{1}{4} = -\frac{10}{12} + \frac{3}{12} = \boxed{-\frac{7}{12}}$

c) $-\frac{3}{4} + (-\frac{2}{5}) = -\frac{15}{20} + (-\frac{8}{20}) = \boxed{-\frac{23}{20}}$ d) $-\frac{3}{8} - \frac{1}{6} = -\frac{9}{24} - \frac{4}{24} = \boxed{-\frac{13}{24}}$

⑧ a) $\frac{5}{6} - (-\frac{5}{9}) = \frac{15}{18} + \frac{10}{18} = \boxed{\frac{25}{18}}$ b) $-\frac{1}{9} - (-\frac{4}{9}) = -\frac{1}{9} + \frac{4}{9} = \frac{3}{9} = \boxed{\frac{1}{3}}$

c) $-\frac{4}{7}(\frac{3}{4}) = \frac{-4 \cdot 3}{7 \cdot 4} = \boxed{-\frac{3}{7}}$ d) $-\frac{1}{4}(-\frac{2}{3}) = \frac{1 \cdot 2}{4 \cdot 3} = \frac{2}{12} = \frac{1}{6} = \boxed{\frac{1}{6}}$

⑨ a) $\frac{3}{8} \div (-\frac{5}{4}) = \frac{3}{8}(-\frac{4}{5}) = -\frac{3 \cdot 4}{8 \cdot 5} = -\frac{3 \cdot 4}{2 \cdot 4 \cdot 5} = \boxed{-\frac{3}{10}}$ b) $-\frac{2}{3} \div (-\frac{4}{5}) = -\frac{2}{3}(-\frac{5}{4}) = \frac{2 \cdot 5}{3 \cdot 4} = \frac{2 \cdot 5}{3 \cdot 2 \cdot 2} = \frac{5}{6} = \boxed{\frac{5}{6}}$
 c) $-5 + (-3) + (-1) = \boxed{-9}$ d) $-3(4)(-2) = -12(-2) = \boxed{24}$

⑩ a) $-4 + 2 - 8 = -2 - 8 = \boxed{-10}$ b) $1 - 3 + 6 = -2 + 6 = \boxed{4}$

c) $-6 - 9 - 4 = -15 - 4 = \boxed{-19}$ d) $2 + (-10) + (-1) + 18 = -8 + (-1) + 18 = -9 + 18 = \boxed{9}$

⑪ a) $-9(-1)(-9) = 9(-9) = \boxed{-81}$ b) $4(2)(-8) = 8(-8) = \boxed{-64}$

c) $-\frac{4}{5} + \frac{1}{10} - \frac{2}{5} = -\frac{6}{5} + \frac{1}{10} = \frac{-12}{10} + \frac{1}{10} = \boxed{-\frac{11}{10}}$ d) $-\frac{1}{4} - \frac{3}{8} + \frac{1}{2} = -\frac{2}{8} - \frac{3}{8} + \frac{4}{8} = -\frac{5}{8} + \frac{4}{8} = \boxed{-\frac{1}{8}}$

$$⑫ a) \frac{2}{5} \left(-\frac{1}{4}\right) \left(\frac{3}{4}\right) = -\frac{2 \cdot 1 \cdot 3}{5 \cdot 4 \cdot 4} = -\frac{2 \cdot 1 \cdot 3}{5 \cdot 2 \cdot 2 \cdot 4} = \boxed{-\frac{3}{40}}$$

$$b) -\frac{1}{6} - \frac{8}{9} - \frac{2}{3} - 1 = -\frac{3}{18} - \frac{16}{18} - \frac{12}{18} - \frac{18}{18} = \boxed{-\frac{49}{18}}$$

$$c) -0.23(-4.6) = \boxed{1.058}$$

$$d) 56.7 + (-3.3) = \boxed{53.4}$$

$$⑬ a) -5x + 6x = 1x = \boxed{x}$$

$$b) 9y - 11y = \boxed{-2y}$$

$$c) -6z - 4z = \boxed{-10z}$$

$$d) -14x + 8x = \boxed{-6x}$$

$$⑭ a) 6x - (-10x) = 6x + 10x = \boxed{16x} \quad b) -3y - (-5y) = -3y + 5y = \boxed{2y}$$

$$c) 2z(-3z) = \boxed{-6z^2} \quad d) \frac{16x^3}{8x} = \boxed{2x^2}$$

$$⑮ a) -3x^4yz^3(-9xyz^5) = \boxed{27x^5y^2z^8}$$

$$b) 5x - 10x - 11x = -5x - 11x = \boxed{-16x}$$

$$⑯ a) 20x^2y(-y)(4z^2y^2x) = (-20x^2y^2)(4z^2y^2x) = \boxed{-80x^3y^4z^2}$$

$$b) \frac{-63x^5yz^7}{9x^7y^2z} = \boxed{-\frac{7z^6}{x^2y}}$$

$$⑰ a) \frac{-20a^3b^4c^2}{-25a^4b^2c^2} = \boxed{\frac{4b^2}{5a}} \quad b) -8x^{10}y^6(5x^3y^4z) = \boxed{-40x^{13}y^{10}z}$$

$$⑱ a) \frac{9x^5y^3z}{-36xyz^4} = \boxed{-\frac{x^4y^2}{4z^3}} \quad b) \frac{8x^{10}y}{32x^{15}y^4} = \boxed{\frac{1}{4x^5y^3}}$$

$$⑲ a) \frac{x}{7} - \frac{2x}{7} = \frac{-1x}{7} = \boxed{-\frac{x}{7}} \quad b) -\frac{3y}{8} + \frac{y}{4} = -\frac{3y}{8} + \frac{2y}{8} = \boxed{-\frac{y}{8}}$$

$$⑳ a) -x + 4x - 3x = 3x - 3x = \boxed{0}$$

$$b) -8y - 9y - 10y + y = -17y - 10y + y = -27y + y = \boxed{-26y}$$

$$\begin{aligned} ㉑) -\frac{5x}{6y^2z} \left(\frac{3x^2yz}{4}\right) &= -\frac{5 \cdot 3x^3yz}{6 \cdot 4y^2z} \\ &= -\frac{5 \cdot 3x^3}{2 \cdot 3 \cdot 4y} \\ &= \boxed{-\frac{5x^3}{8y}} \end{aligned}$$

$$\begin{aligned}
 (22) \quad \frac{7x^4y^3z^6}{3} \div \left(-\frac{5x^3y^3}{6z^4}\right) &= \frac{7x^4y^3z^6}{3} \left(-\frac{6z^4}{5x^3y^3}\right) \\
 &= \frac{-7 \cdot 6 x^4 y^3 z^{10}}{3 \cdot 5 x^3 y^3} = \frac{-7 \cdot 2 \cdot 3 \cancel{x} z^{10}}{3 \cdot 5} \\
 &= \boxed{-\frac{14xz^{10}}{5}}
 \end{aligned}$$

$$(23) \quad a) -\frac{3x}{4} + \frac{x}{9} + 2x - \left(-\frac{2x}{3}\right) \quad b) -2y + (2y)$$

$$-\frac{27x}{36} + \frac{4x}{36} + \frac{72x}{36} + \frac{24x}{36}$$

$$-\frac{23x}{36} + \frac{72x}{36} + \frac{24x}{36}$$

$$\frac{49x}{36} + \frac{24x}{36}$$

$$\boxed{\frac{73x}{36}}$$

$$\begin{aligned}
 (24) \quad -\frac{2a^2b^3}{5c} \div (-3b^2c) &= -\frac{2a^2b^3}{5c} \left(\frac{1}{-3b^2c}\right) \\
 &= \frac{-2a^2b^3}{-15b^2c^2} = \boxed{\frac{2a^2b}{15c^2}}
 \end{aligned}$$

$$\begin{aligned}
 (25) \quad (-5x^7y^3z^8) \left(\frac{9x^5y^3}{10z^7}\right) &= \frac{(-5x^7y^3z^8)}{1} \left(\frac{9x^5y^3}{10z^7}\right) \\
 &= \frac{-5 \cdot 9 x^{12} y^6 z^8}{10z^7} \\
 &= -\frac{5 \cdot 9 x^{12} y^6 \cancel{z}}{2 \cdot 5} = \boxed{-\frac{9x^{12}y^6z}{2}}
 \end{aligned}$$